KAIZEN MATHS

Lesson starter

Work, Energy and Power

Formula Basics

Name _____

S1

A constant force of 60 N moves an object 4 m in the direction of the force. Find the work done.

S2

A mass of 10 kg is raised vertically by 1.5 m. Find the gain in gravitational potential energy. Take g = 9.8 m s^{-2}.

S3

A body of mass 3 kg moves at 4 m s^{-1}. Find its kinetic energy.

S4

900 J of work is done in 2 s. Find the power.

Show your working

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MEC-8E86-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Work, Energy and Power

Formula Basics

Name _____

E1

A constant force of 80 N moves an object 40 m in the direction of the force. Find the work done.

E2

A mass of 12 kg is raised vertically by 5 m. Find the gain in gravitational potential energy. Take g = 9.8 m s^{-2}.

E3

A body of mass 10 kg moves at 15 m s^{-1}. Find its kinetic energy.

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MEC-8E86-L1 · E

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Show your working

Teacher answers & working

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S1

A constant force of 60 N moves an object 4 m in the direction of the force. Find the work done.

Answer:

240 J

Working:

Work done equals force multiplied by distance moved in the direction of the force.

$W = Fs$

$F = 60, s = 4$

$W = 60 \times 4$

$W = 240\ J$

S2

A mass of 10 kg is raised vertically by 1.5 m. Find the gain in gravitational potential energy. Take g = 9.8 m s^{-2}.

Answer:

147 J

Working:

Gain in gravitational potential energy is mgh.

$GPE = mgh$

$m = 10, g = 9.8, h = 1.5$

$GPE = 10 \times 9.8 \times 1.5$

$GPE = 147\ J$

S3

A body of mass 3 kg moves at 4 m s^{-1}. Find its kinetic energy.

Answer:

24 J

Working:

Kinetic energy is one half times mass times speed squared.

$KE = \frac{1}{2}mv^2$

$m = 3, v = 4$

$KE = \frac{1}{2} \times 3 \times 4^2$

$KE = \frac{1}{2} \times 3 \times 16$

$KE = 24\ J$

Teacher answers & working

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S4

900 J of work is done in 2 s. Find the power.

Answer:

450 W

Working:

Power is work done divided by time taken.

$P = \frac{W}{t}$

$W = 900, t = 2$

$P = \frac{900}{2}$

$P = 450$ W

E1

A constant force of 80 N moves an object 40 m in the direction of the force. Find the work done.

Answer:

3200 J

Working:

Work done equals force multiplied by distance moved in the direction of the force.

$W = Fs$

$F = 80, s = 40$

$W = 80 \times 40$

$W = 3200$ J

E2

A mass of 12 kg is raised vertically by 5 m. Find the gain in gravitational potential energy. Take g = 9.8 m s^{-2}.

Answer:

588 J

Working:

Gain in gravitational potential energy is mgh.

$GPE = mgh$

$m = 12, g = 9.8, h = 5$

$GPE = 12 \times 9.8 \times 5$

$GPE = 588$ J

Teacher answers & working

Work, Energy and Power · Formula Basics

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E3

A body of mass 10 kg moves at 15 m s^{-1}. Find its kinetic energy.

Answer:

1125 J

Working:

Kinetic energy is one half times mass times speed squared.

$\mathrm{KE} = \frac{1}{2}\mathrm{mv}^2$

$\mathrm{m} = 10, \mathrm{v} = 15$

$\mathrm{KE} = \frac{1}{2} \times 10 \times 15^2$

$\mathrm{KE} = \frac{1}{2} \times 10 \times 225$

$\mathrm{KE} = 1125\ \mathrm{J}$